

Burst-Controlled Phantom-Queue Policer in eBPF

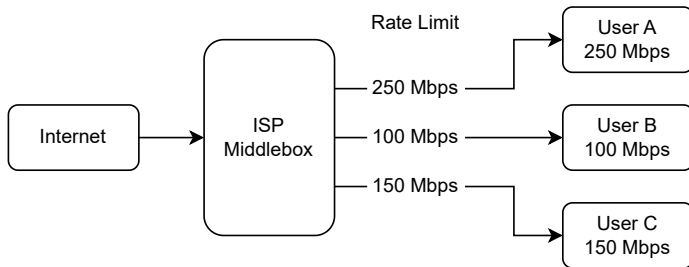
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Internet and Networked Systems Project
supervised by Jonas Köppeler

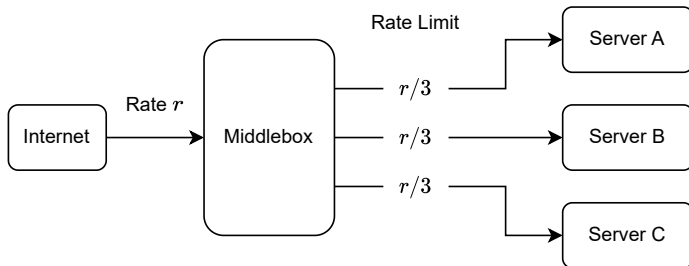
Rate limiting use cases

- Rate limiting



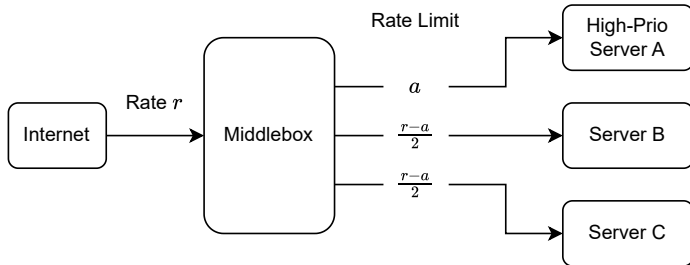
Rate limiting use cases

- Rate limiting
- Fairness

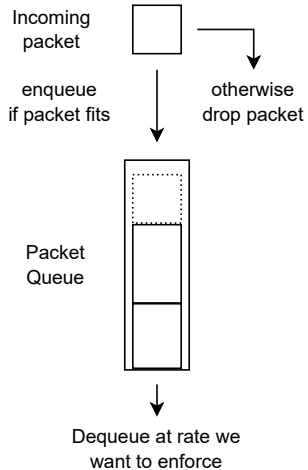


Rate limiting use cases

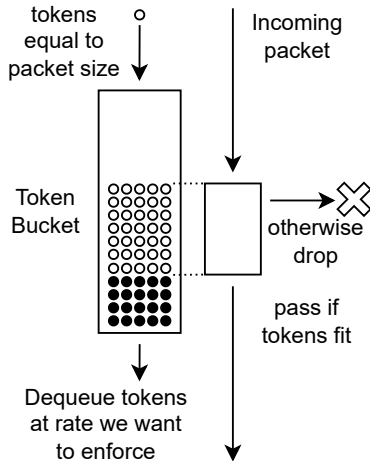
- Rate limiting
- Fairness
- Prioritisation

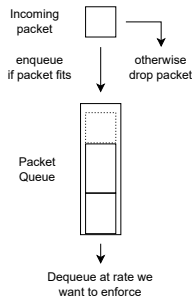


Traffic Shaper



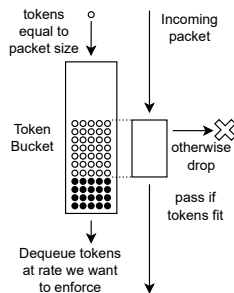
Traffic Policer





(a) Traffic Shaper

- very accurate rate enforcement
- packet delay
- computationally inefficient and expensive



(b) Traffic Policer

- light-weight
- poor rate enforcement

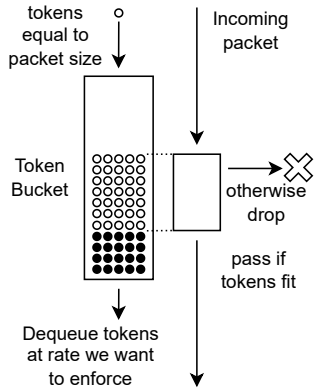
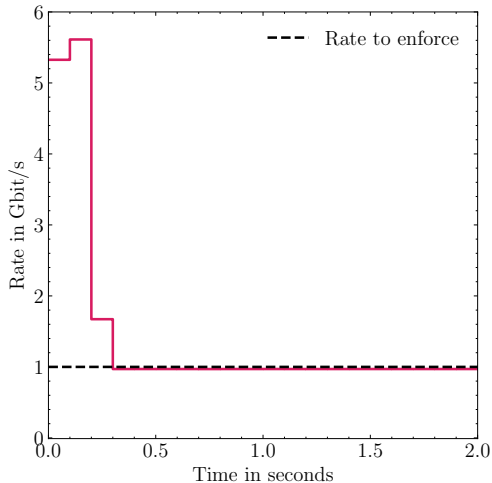


Figure 2: Traffic Policer



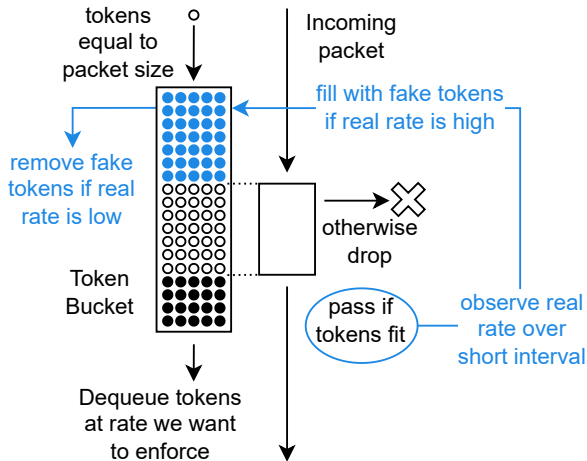


Figure 3: Burst-controlled phantom-queue policer (BC-PQP) [1]

[1] Tahir et al. “Efficient Policy-Rich Rate Enforcement with Phantom Queues”

Limitations of BC-PQP Reference Implementation

implemented in Intel's DPDK:

- single-core implementation
- limited portability
- separated from OS

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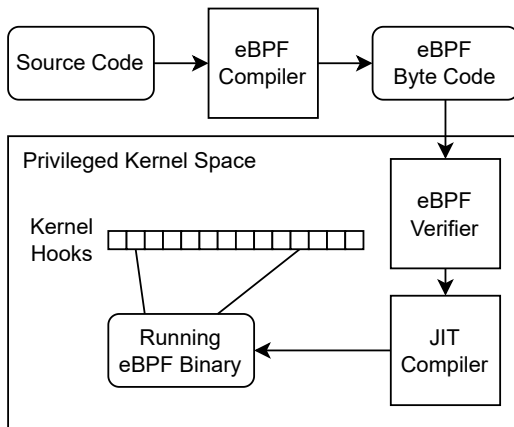
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Our Goal: implement BC-PQP in eBPF

Tool to extend Linux kernel

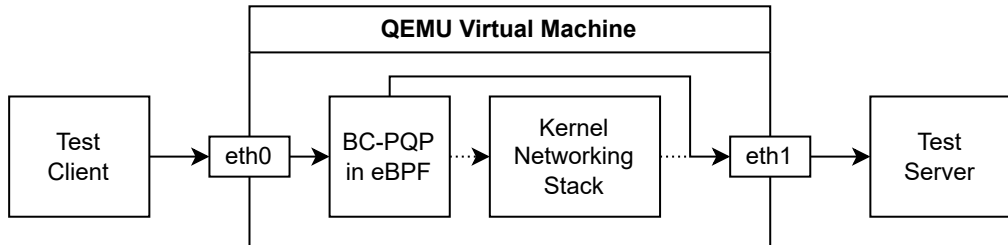
Tool to extend Linux kernel



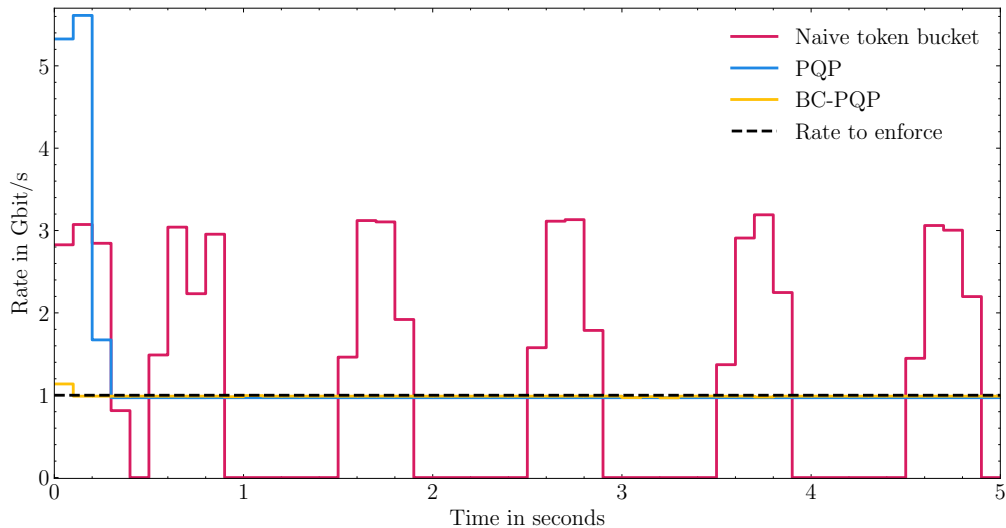
BC-PQP implementation in eBPF

- portability
- multiple cores
- integration with Linux networking stack
- application safety

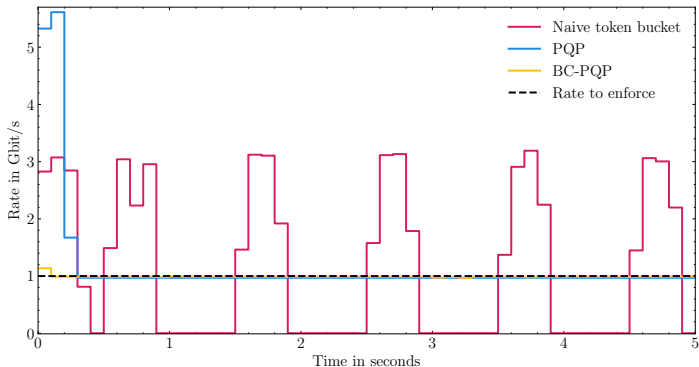
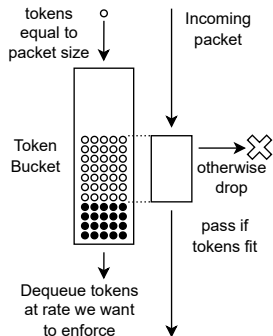
Progress: Test Setup



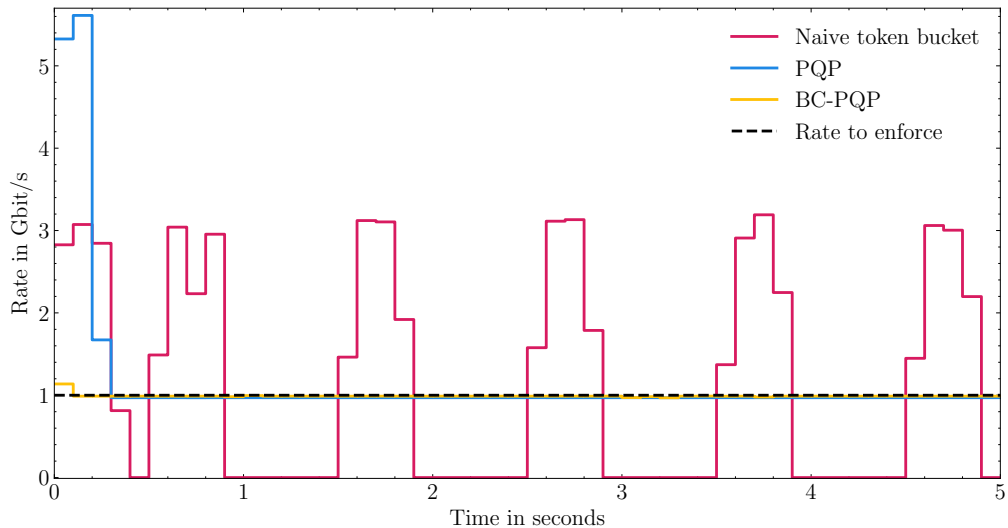
Progress: Implementation



Progress: Implementation



Progress: Implementation



Conclusion

Current state:

- fully functional BC-PQP implementation¹
- multi-core support

¹<https://github.com/ftschirpke/BC-PQP-eBPF>

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Outlook:

- improve multi-core scaling
- full evaluation regarding throughput, fairness, and scaling

¹<https://github.com/ftschirpke/BC-PQP-eBPF>

- [1] A. Tahir, P. Goyal, I. Marinos, M. Evans, and R. Mittal, **“Efficient Policy-Rich Rate Enforcement with Phantom Queues”**, in *Proceedings of the ACM SIGCOMM 2024 Conference*, Sydney NSW Australia: ACM, 2024, pp. 1000–1013.
- [2] S. Kunniyur and R. Srikant, **“An Adaptive Virtual Queue (AVQ) Algorithm for Active Queue Management”**, *IEEE/ACM Transactions on Networking*, vol. 12, no. 2, pp. 286–299, 2004.